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GRADUATE SCHOOL



MASTER SPACE 1
INTERNSHIP REPORT

Project:

**UPGRADING OF THE TELESCOPE SYSTEM AND
SOME RESULTS AT NHA TRANG OBSERVATORY**

Student:	La Thuy Linh
Academic year:	2021-2022
External Supervisor:	Dr. Nguyen Thi Thao
Host Institute:	Nha Trang Observatory, Vietnam National Space Center
Internal Supervisor:	Dr. Phan Thanh Hien (USTH)

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Abbreviations

Advanced Control System	ACS
Charge Coupled Device	CCD
Field of View	FOV
Finger Lakes Intrumentation	FLI
C/2017 K2 PanSTARRS Comet	K2
Nha Trang Observatory	NTO
Quantum Efficiency	QE
Ritchey-Chrétien 500 mm	RC500
Red, Green, Blue (combination)	RGB
Screen Transfer Function	STF
Telescope Control System	TCS
Vietnam National Space Center	VNSC

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Abstract

Nha Trang Observatory host a 50-cm (f/8) Ritchey - Chrétien reflecting telescope and a Charged-Couple Device (CCD) at its focal plane. Recently, the system has been equipped with a new filter set including H-alpha, S-II, and O-III and upgraded the monitoring and control operation into an open and free software, namely Nighttime Imaging 'N' Astronomy (N.I.N.A) in order to combine a weather station with the former system. The CCD array was contaminated by water vapor and we are working to recover it. The images taken with this telescope system were processed and displayed in this report.

1 Introduction to Nha Trang Observatory Telescope System

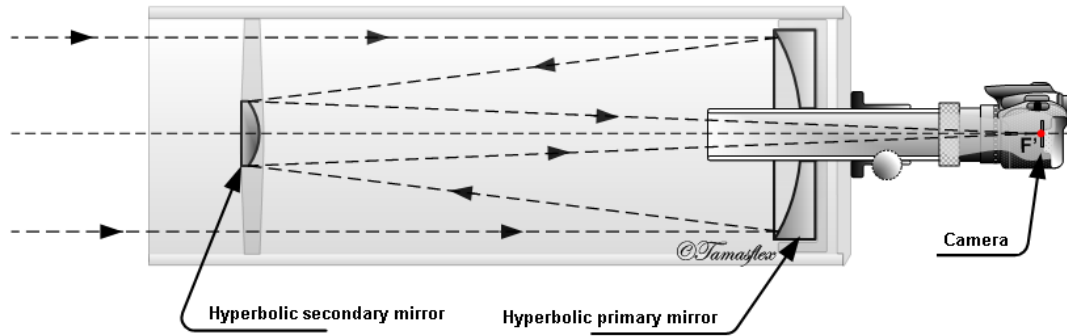
Nha Trang Observatory (NTO) belongs to Vietnam National Space Center (VNSC), located in Hon Chong Hill, Nha Trang, Vietnam (Latitude: 12.272° North, Longitude: 109.205° East). To the North of NTO is urban area and the beach, which is lighted up all night, meanwhile the South of NTO faces the sea and not heavily lighted. The sea atmosphere has high humidity and the temperature is around 24 to 30°C . During the day, the sky is usually clear and sunny, but the clouds will form in the late afternoon and night, hence the observing time is quite limited.



Figure 1: The RC500 Telescope in NTO

NTO is equipped with a 50-cm Ritchey-Chrétien (RC500) reflecting telescope system, and operates in synchronization with an 8-meter dome, as shown in Figure 1. The telescope is mounted on an altitude-azimuth mount, and can point to nearly every position in the sky. The Ritchey-Chrétien has the configuration as in Figure 2 of the Cassegrain telescope system, which produces no spherical aberration or coma [**to measure the sky**]. My objective in this internship is to learn about operation of the telescope system and its apparatus, which will be clearly described in Section 1. In addition, I assisted to upgrade the telescope system with new filters and softwares, depicted in Section 2, and processed the images that were taken with this telescope with the procedure interpreted in Section

3. The outcome of this process are the Ring Nebula image and C/2017 K2 Comet image, illustrated in Section 4.



Ritchey - Chrétien (RCT)

Figure 2: Ritchey-Chrétien telescope configuration [RCconfig]

When the system is ready (i.e. aligned and calibrated) for object pointing or slewing, all the actions are controlled by the computer. Figure 3 indicates the link between the soft and hard wares. The softwares are indicated in yellow and cyan color, while the hardwares are indicated in pink. All these softwares communicate through ASCOM, which is an interface for the astronomical softwares and devices to communicate smoothly with each other [ascom]. The SkyX Professional is the main software utilized for processing, observing, and taking images because it can communicate with other softwares and controls the equipment via these softwares. During operating sessions, sometimes the system encounters soft limits (the line 0° and 360°) and hard limits (when the cables are tense due to the rotation of the telescope), then it needs to be driven manually, through the computer or by the hand controller.

- CBDome (short for Columbia Dome): Control the rotation and shutter of the dome. The full opening of the dome takes 1.5 minutes and the full rotation of the dome takes several minutes.
- MAESTRO: A specific software that controls the Telescope Control System (TCS), in other words, controls the altitudinal and azimuthal motors.
- ATC Remote or ScopeFocuser: These softwares are for changing the focus point of the telescope via changing the position of the secondary mirror. ATC Remote is the one that is included in the system when the system was first purchased, however, it does not have a feedback loop and does not seem to recognize its real position due

to lack of encoder, thus another option for this is the ScopeFocuser, accompanied with one Arduino board controller, which monitors the secondary mirror motor and it implements the lacks of the ATC Remote.

- Pyxis: Controls the field rotator, which implements the rotation of the sky. This is very necessary for long-time exposure frames, since it compensate for the rotation of the sky and prevent the trails of the stars while taking the images.
- PHD2 Guiding: This software goes with a guiding camera and helps track the object at the correct rate. This software communicates directly to the TCS and can rotate the altazimuth motors. In normal conditions, the MAESTRO and the PHD2 Guiding do not conflict.

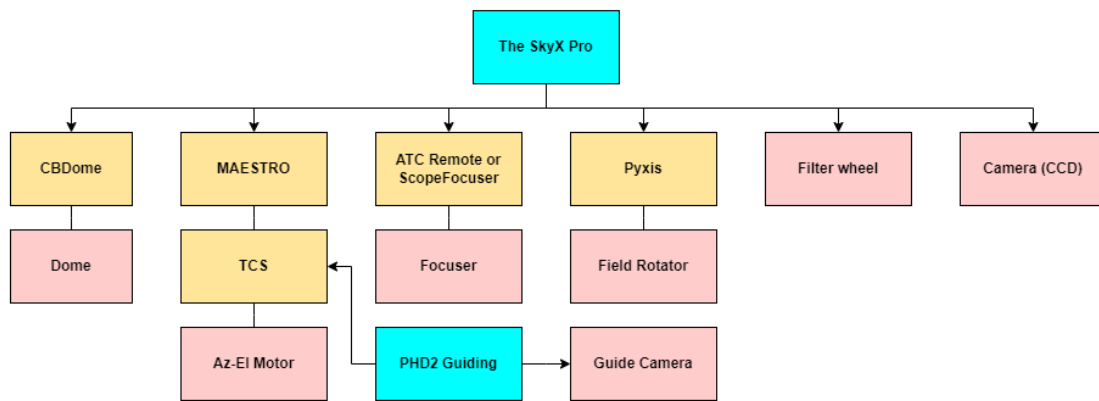


Figure 3: Telescope System

Figure 4 shows the pictures of the equipment that are indicated in the telescope system. These apparatus are coincidental and the images are focused on the CCD plane. The detailed setup of the imaging system is pictured in Figure 6.

- Telescope Tube: 50-cm Ritchey-Chretien Reflecting Telescope, f/8.
- Telescope Control System: the base of the telescope, helps to rotate the telescope system. It can rotate full 360 degrees in 2 to 5 minutes.
- The Pyxis Camera field rotator: compensates for the rotation of the sky, hence the images will stay in the right position for long exposure. Roughly saying, this equipment is used for changing the local horizontal coordinate to the equatorial coordinate.

- Filter wheel FLI CFW3-10: This is a 10-slot filter wheel, and will be discussed in more detail in section 2.
- CCD FLI ProLine PL16801: The main equipment that served the purpose of taking the images. Further details of this CCD can be found in [pl16801].

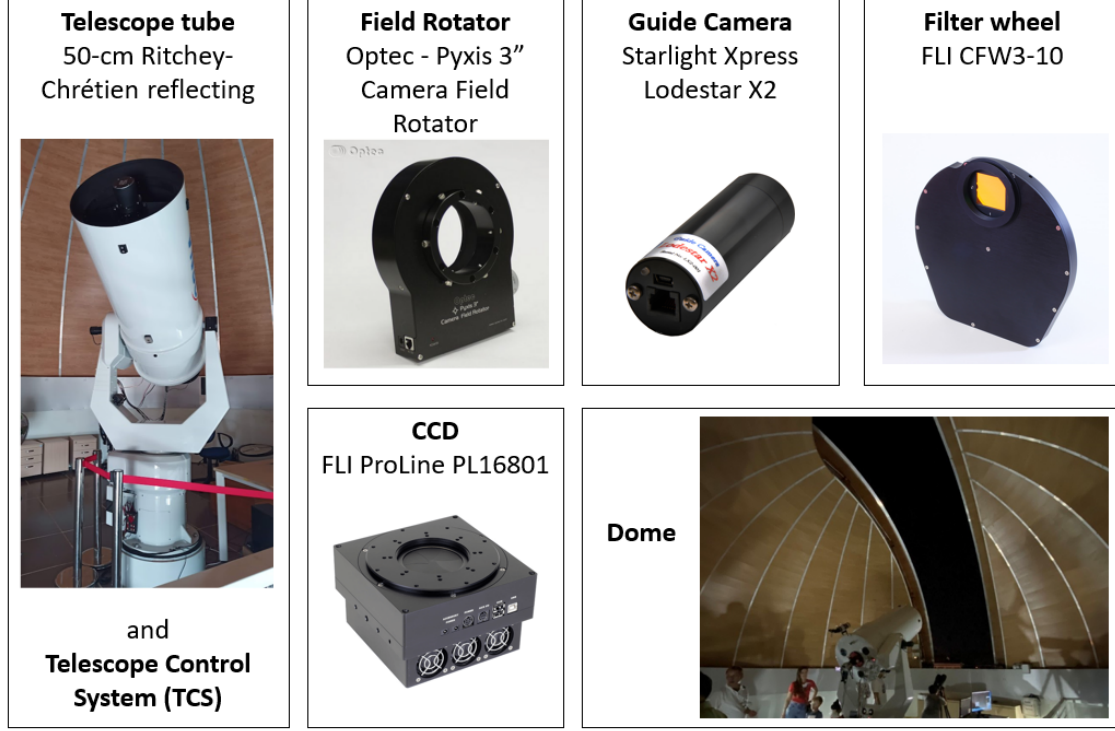


Figure 4: Equipment

The field of view (FOV) of a telescope imaging system using CCD can be calculated by the formula $\theta = \arctan(D \times 10^{-3}/f)$, where θ is the FOV in one direction in arcminute, D is the length (in mm) of the CCD in that direction, and f is the focal length of the telescope in mm. The length of the CCD in one direction is (pixel size) $\times$ (number of pixel in one direction) = $9 \mu\text{m} \times 4096 = 36.864 \text{ mm}$. With the focal length of $f = 4000 \text{ mm}$, the FOV in one direction is 0.528 degree. The summary of parameters of the imaging system is shown in Figure 5 and Table 1. Given the FOV of the imaging system, we know if object lies entirely inside the images (e.g. the M57, M20, etc.) or not (e.g. M31). If the object size is larger than the FOV, we need to mosaic the images so that we can capture the whole object.

Choose Equipment Add New Equipment To Database

Telescope: Focal Length: mm Aperture: mm

Camera: Resolution: x px Pixel Size: x μm

Barlow / Reducer: Binning: Angle:

Focal Ratio: Resolution: Field of View: Dawes Limit: Add to View

Figure 5: FOV calculation from [fovcal]

Table 1: Main parameters of the system when combined

Telescope			Sensor			Guiding system	Picture saving time after exposure (s)	FOV	FOV 1 pixel (deg)	Owner	Location
Name	Focus (mm)	Diameter (mm)	Camera	Resolution	Pixel Size (um)	Maximum rotation rate(deg/s)					
RC500	4000	500	FLI - PL16801	4096 x 4096	9	0.5	4 seconds after exposure	0.53° x 0.53°	0.00013	VNSC, Nha Trang	12.272 N, 109.205 E

2 Upgrading of Nha Trang Observatory Telescope System

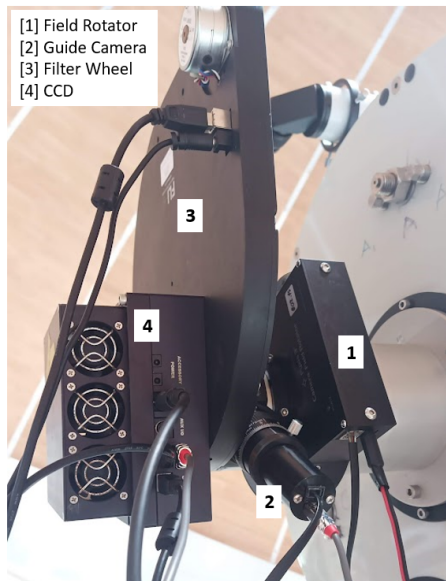


Figure 6: Imaging system setup



Figure 7: Inside the Filter wheel

The imaging system is placed behind the telescope tube, all the equipment is placed co-incidentally with the order as shown in Figure 6. The light from the tube will go straight through the field rotator (the field rotator does not affect the propagation of light), then a small fraction of light will be reflected by a small diagonal mirror to go to the guide camera. Rest of the light will go through the selected filter in the filter wheel and then reach the CCD.

2.1 Adding S-II, O-III, and H α filters

The FLI CFW3-10 Filter Wheel has 10 slots for inserting the filters. One of these slots will usually be set blank, for taking the frames other than light frames (e.g flat field frames), and this slot is indicated as *slot0* in SkyX. Because of the longitudinal transverse aberration and the difference in thickness of the filter, it requires the secondary mirror to move longitudinally inside the telescope tube so that the images can be focused precisely on the CCD array. The ScopeFocuser, in replacement for ATC Remote, is in charge of finding the right position of the secondary mirror. The SkyX Pro recognizes and controls the ScopeFocuser by its built-in function *@Focus2 Runs*. The SkyX Pro will continuously take images of bright stars and moves the secondary mirror to the position where the star is well focused in the CCD plane. During this session, more than 20 frames will be taken with a short exposure and large binning, since we do not require high resolution for these sample frames. Figure 8 depicts the procedure of taking the image and calibrating it and Figure 9 shows some of the images that are taken during the focus finding session.

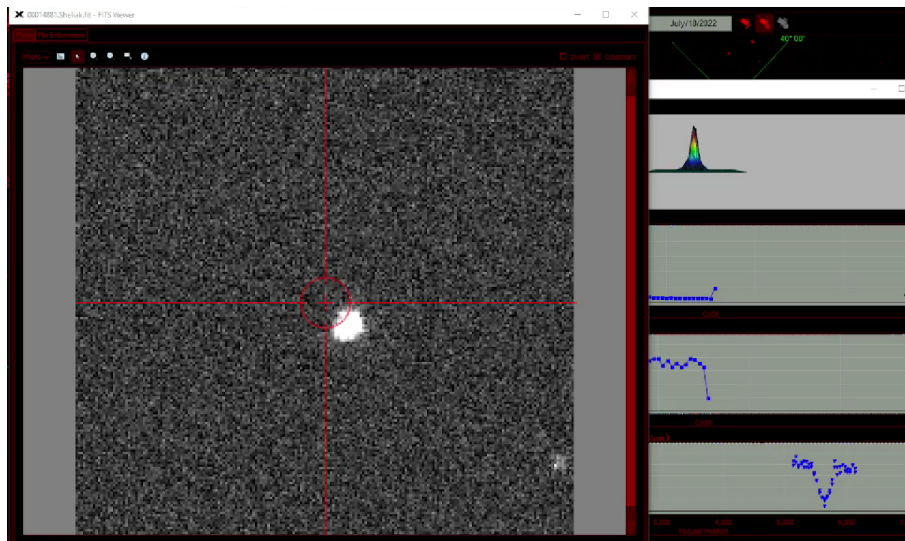


Figure 8: @Focus2 Runs

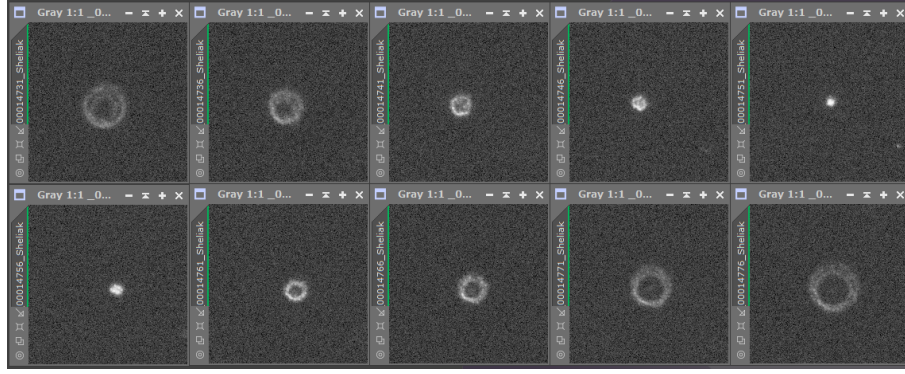


Figure 9: @Focus2 Runs sample frames

Figure 7 is the image of the filter wheel and how the filters are put inside the filter wheel, and Figure 10 are the new set of filters that we used for imaging, including the $H\alpha$, O-III and S-II filters. These filters are manufactured by Finger Lakes Instrumentation Inc., hence they share similar characteristics: IR blocked (for blocking the infra-red spectrum, without modifying the visible spectrum) [**IR'block**], Anti-Reflection (AR) Coated (increasing the throughput of the optical system and reduce the wrong signal caused by reflections) [**AR'coat**], plane-optical polished, hard coated [**hard'coat'1**, **hard'coat'2**] and sealed coating edge. They have the size of 50 mm x 50 mm, but the bandwidth is different from each other.

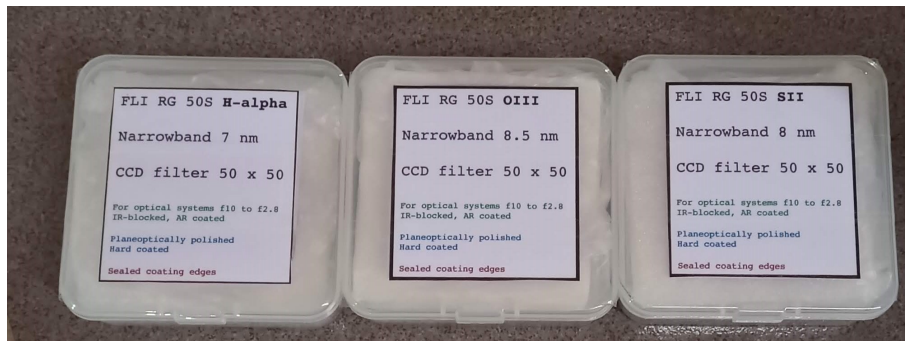


Figure 10: $H\alpha$, O-III and S-II filter

H-alpha ($H\alpha$) [**h-alpha**] (wavelength 656.281 nm) is a deep-red visible spectral line in the Balmer series, that occurs when the electron of a hydrogen atom falls from the third lowest energy level to the second. H-alpha spectral line is obtained when used for observing the solar prominences and the chromosphere of the Sun, and the nebulas. The O-III [**o-iii**] (wavelength 495.9 and 500.7 nm) and S-II [**sulfur**] (wavelength 672.4 nm) are also frequently observed in the nebulas. The color combination of these three channels is also a popular combination in astrophotography. The H-alpha has the deep-red color,

while O-III has the green-ish color and the S-II will be assigned to the blue color, even though it lies in the red spectrum. Besides the new filters, we still exploit other filters that are available in NTO, such as the LRGB filter set.

Table 2: Main parameters of the filters

	H-alpha	O-III	S-II
Size (mm x mm)	50 x 50	50 x 50	50 x 50
IR block	✓	✓	✓
AR coated	✓	✓	✓
Planeoptically polished	✓	✓	✓
Hard coated	✓	✓	✓
Sealed coating edge	✓	✓	✓
Half Bandwidth (nm)	7	8.5	8
Peak (nm)	656.3	502	673
Focuser Position	5629	5635	5641

2.2 Automatism of the system

Since the weather condition in NTO is not ideal, with the cloudy and humid conditions, NTO plans to install a full-sky camera and a weather station (including wind sensor, temperature sensor, and humidity sensor), in order to examine the weather at NTO and decide at what time can we take the photos.

Even though SkyX Professional is good at monitoring all the other softwares, it is not considered a good choice to integrate the weather detection plug-in. Hence, another option instead of the SkyX Pro is N.I.N.A (Nighttime Imaging 'N' Astronomy) [[nina](#)], a free and open source software that can run the same tasks as SkyX Pro and it has more plug-ins and can connect with other software such as Stellarium [[stellarium](#)] for good pointing and visualization of the sky. However, the catalogs of objects of Stellarium and N.I.N.A are not as good as that of SkyX. For example, when I looked up the C/2017 K2 comet, it was not included in Stellarium's catalog yet, meanwhile, SkyX has the information about this comet; in this case, we must use SkyX for taking the images. The installation of N.I.N.A has been done and the weather station will be soon integrated in the system.

3 Procedure of taking images and image processing

Before combining the light frames to generate colored images, we need to calibrate them. The calibration procedure requires other types of frames to cancel out the noises that the

mirrors and the CCD causes. Image reduction formula:

$$N_{photons} = (Raw - Bias - Dark * t/t_{dark}) / Flat / Gain \quad (1)$$

where t and t_{dark} respectively the exposure time of the light frames and the dark frames. The math is done pixel by pixel.

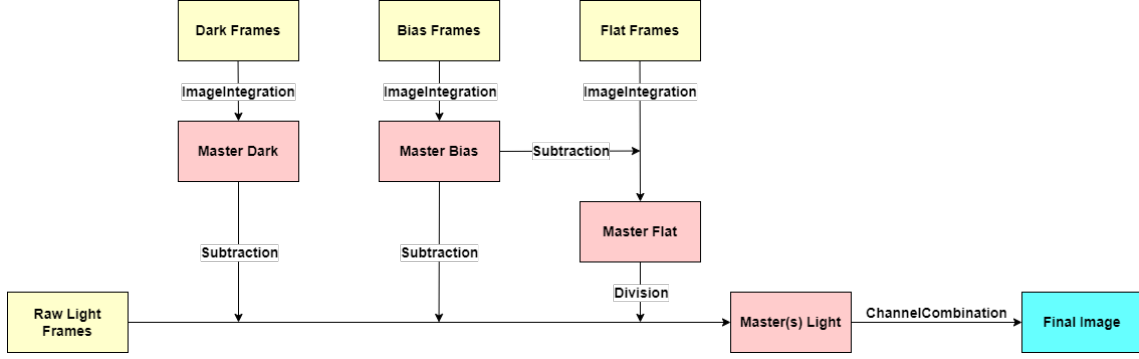


Figure 11: Image Processing Process

All of the images we process in this report were taken by the NTO's telescope system from 8 p.m on July 18th, 2022 to 5 a.m on July 19th, 2022. The light frames were taken at night time and the calibration frames were taken in the morning of July 19th. On that day, the Moon rose at around 11 p.m, thus we preferred the target which is far from the Moon so that the moonlight did not contaminate the light frames. Nonetheless, the city and the beach are in the North of the NTO, and this brings about a lot of light pollution to the sky and will affect the frames heavily, especially when the object is close to the horizon. For image processing, we use PixInsight [[pixinsight](#)], a very popular tool for astrophotography, because it provides a great deal of options for processing algorithms. Equation 1 can be automatically applied to the frames by the *BatchProcessing* provided by PixInsight, or be done step by step. I have done it first using *BatchProcessing*, but the output images are still very noisy, and sometimes it returns errors since some of the images cannot meet the criteria that were automatically defined by the software. Recalculating those output frames will produce more and more noise in the images, not to mention time-consuming. Thus I processed the images step by step, which will be further discussed in subsection 3.2.

3.1 Calibration Frames

The calibration frames are exploited to calibrate the light frames. There are several types of calibration frames, including: dark, bias, flat field, and dark flat frame. Each type of

calibration frame will need a certain number of individual frames, and these raw frames need to be processed to create a master frame for each type. The three types of calibration frames that we took are dark, bias, and flat field frames.

3.1.1 Bias Frames

Bias frames are used as the pedestal for the light frames, taken at zero exposure time (1 ms) with the cap cover (i.e. no exposure to light). The bias frame is the response of the detector when it reads out, which means the detector may produce positive values for some pixels even though there is no real signal reaching the detector. This *read out noise* will be presented in every frame, and need to be subtracted from the light frames [to measure the sky]. With a set of 15 bias frames, I choose the *Average* combination with Winsorized Sigma Clipping, No Normalization for the integration of the bias frames.

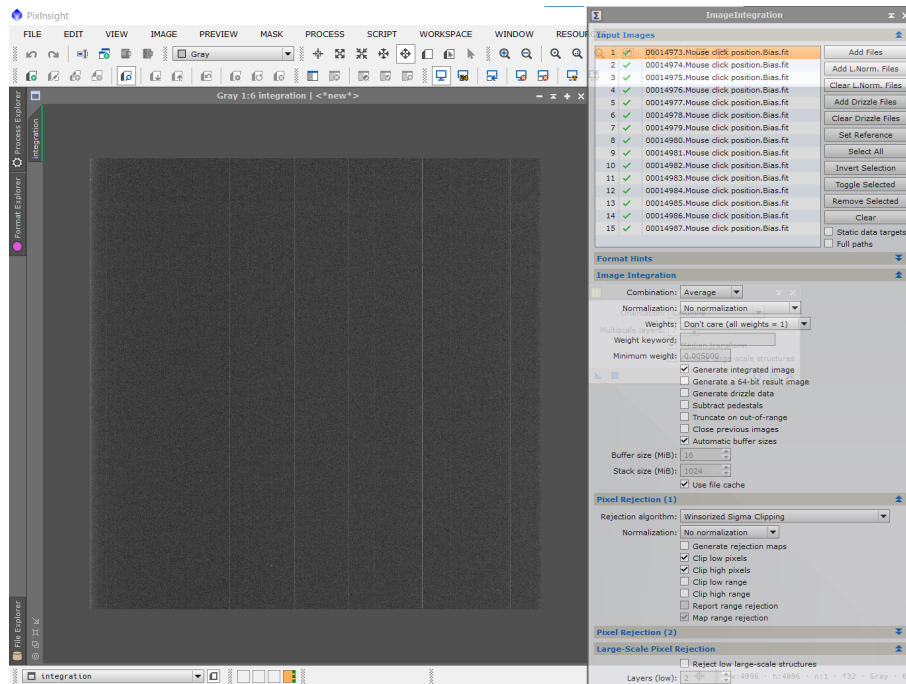


Figure 12: Bias frame integration

After that, we create a Master Bias by applying SuperBias to the integration of bias frames in Figure 12. This step is to produce a noise-free bias frame [superbias], which truly represents the pedestal signal that we require for image calibration.

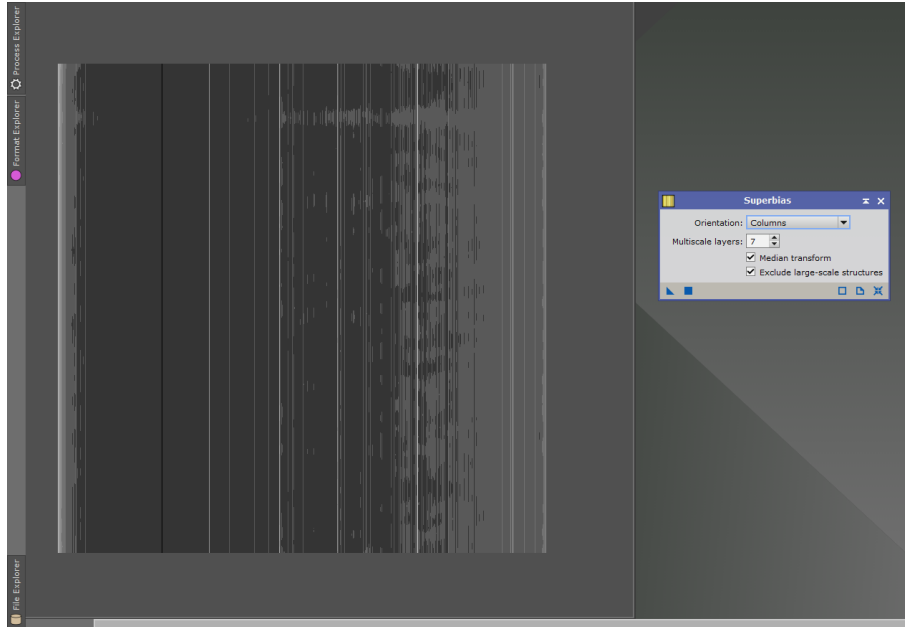


Figure 13: Master Bias

3.1.2 Dark frames

A dark frame is taken with the camera lens (or telescope) covered to ensure there is no light – this means any signal in the dark frame is due to noise, mostly thermal noise (or so-called *dark current* [**to measure the sky**]). The dark frame must be captured under identical conditions (such as exposure, gain, resolution, and temperature) as the light frames so that the noise can be well cancelled. Since we took the light frames in two exposure time (180 seconds and 20 seconds), the 180-second and 20-second dark frames are required, and we took 5 frames for each exposure time. The dark master frame is created by average integration of the individual dark frames with no normalization. Once a dark frame master has been created, it can be subtracted from each image frame to remove much of the noise.

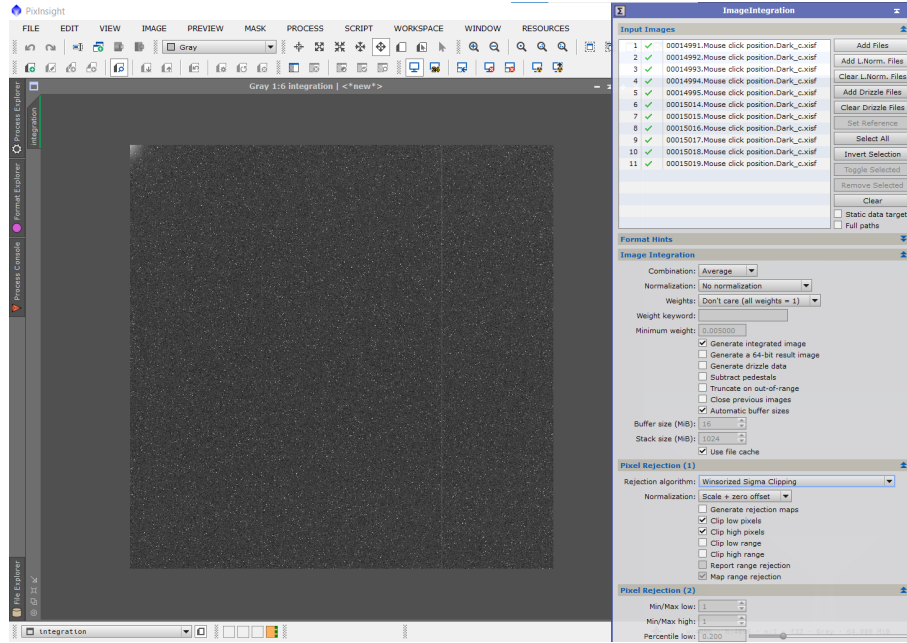


Figure 14: Master dark

3.1.3 Flat field frames

The third type of calibration is flat field frames. They are usually taken against a synthetic white background and help to eliminate the effects caused by dust on the sensor and vignetting. By taking this flat field frame, we can also account for the varying quantum efficiency across the array of pixels. We can take this frame by covering the telescope tube with a white cloth and then illuminate it with a flashlight, or take the image at dawn or dusk when the sky is uniformly illuminated. The SkyX Pro allows us to take a series of flat field frames, then it will automatically take flat frames with different exposure times. In our case, we choose to take 15 flat field frames, and the exposure times vary from 1 second to 10 seconds. This flat field frames was taken at dusk at 5 A.M. on July 19th.

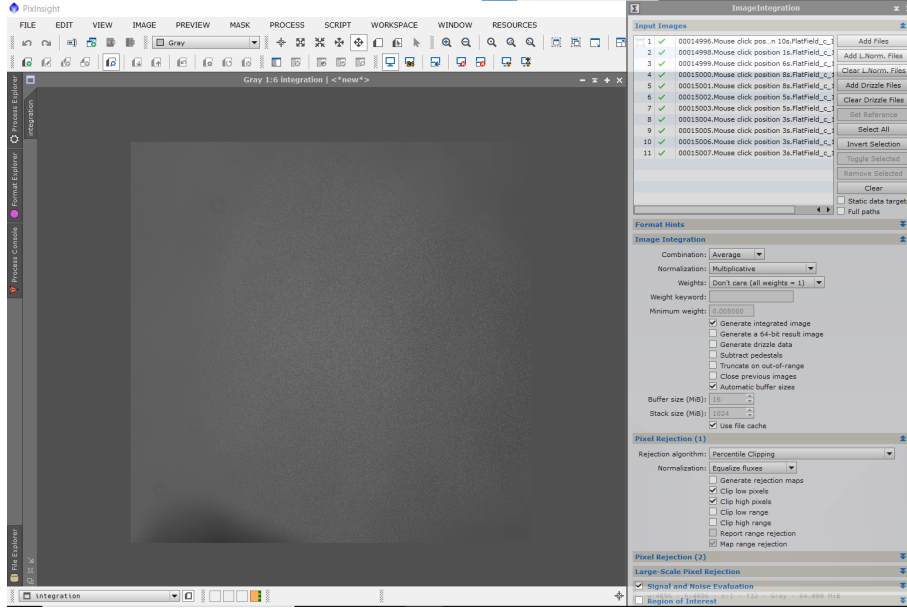


Figure 15: Flat field master frame

A dark region at the left lower edge of the frame, due to the diagonal mirror of the guiding tube, is easily viewed in every light frames and in the flat field frames, as in Figure 15 and 17. In Figure 15, the left hand side and the corners of the images seem to be smoother and darker than the main center of the images.

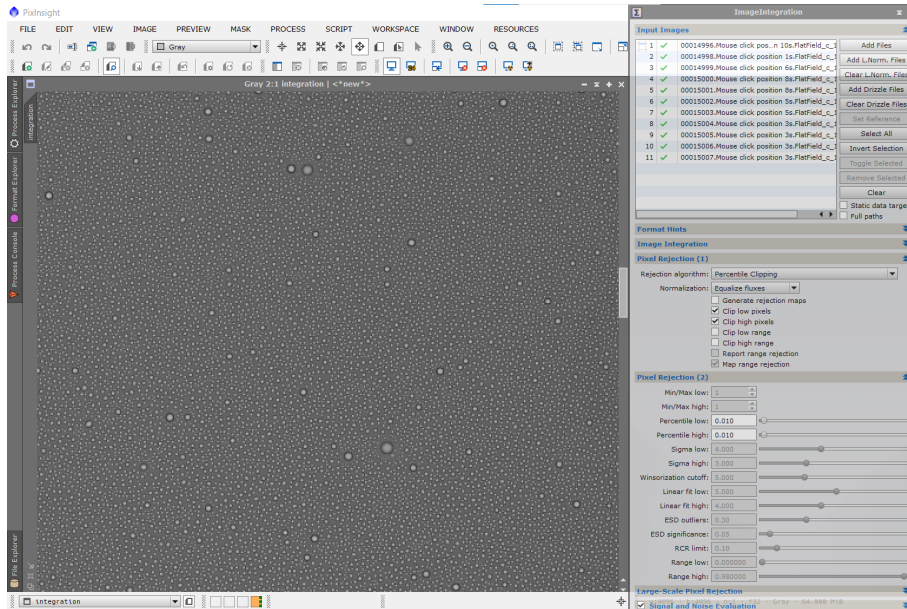


Figure 16: Zoom in flat field frame

The droplet pattern on the CCD array is clearly observed when we zoom in the images, in Figure 16. The droplets distributed roughly on 80% the surface of the array, with

various sizes, ranging from 2 up to 8 pixels in diameter. The appearance of droplets causes erratic pattern and such pattern will significantly harm the light frames. The dust on any reflecting surface in the system also results in the images as faint dark large ring like in Figure 17, but this effect will be effectively canceled out.

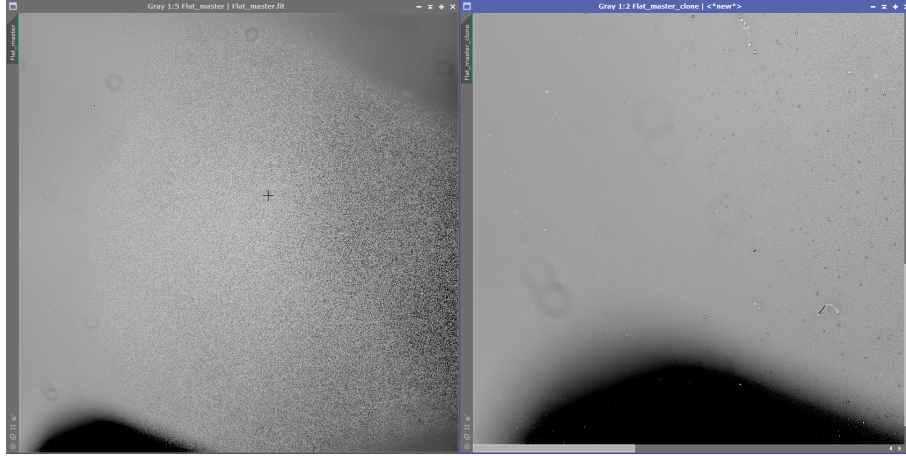


Figure 17: Dust on the mirror

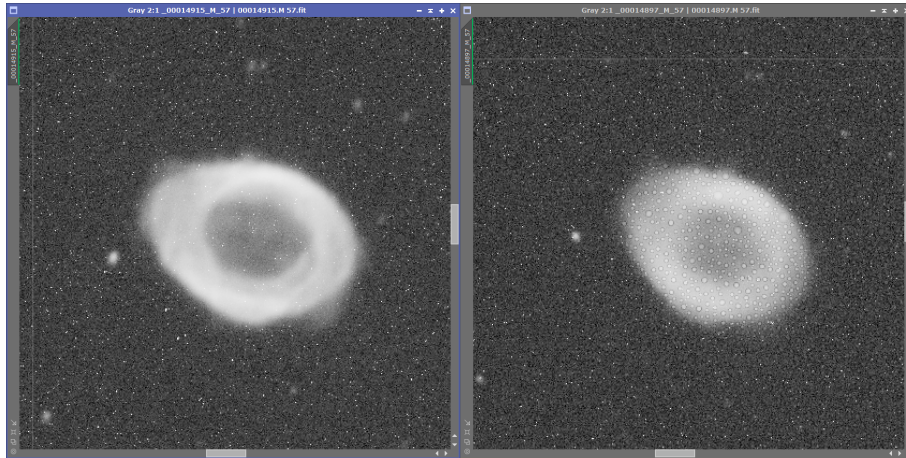


Figure 18: Two raw images of M57 taken in different temperature, left: 10 Celcius degrees, right: 5 Celcius degrees

There is one questionable phenomena that I have not figured out yet is that the light frames that were taken at 10°C do not have the droplets pattern on the CCD surface, meanwhile the flat field frames taken at the same temperature do. During the imaging session, we also tried to heat up and cooled down the CCD several times to see whether we still see the droplets, so that the droplet pattern is different each time. We did not take the flat field at the very moment that we take the light frame, thus the droplet pattern on the CCD of the flat field frames and the light frames are different (I tried using the

flat field frame that we took but not only it did not cancel out but it went worse). There is a trade-off between the thermal noise and the droplet pattern. Due to this problem, when doing further processing, I did not use the flat field frames and I used the 10°C light frames, even though it is very noisy and the water content in the CCD chamber may affect it.

3.2 Light frames

3.2.1 Calibration

Using equation 1, we apply the master frames (bias, flat, and dark) to the light frames to calibrate these light frames before further modification. PixInsight automatically perform this operation, with the master input, as depicted in Figure 19.

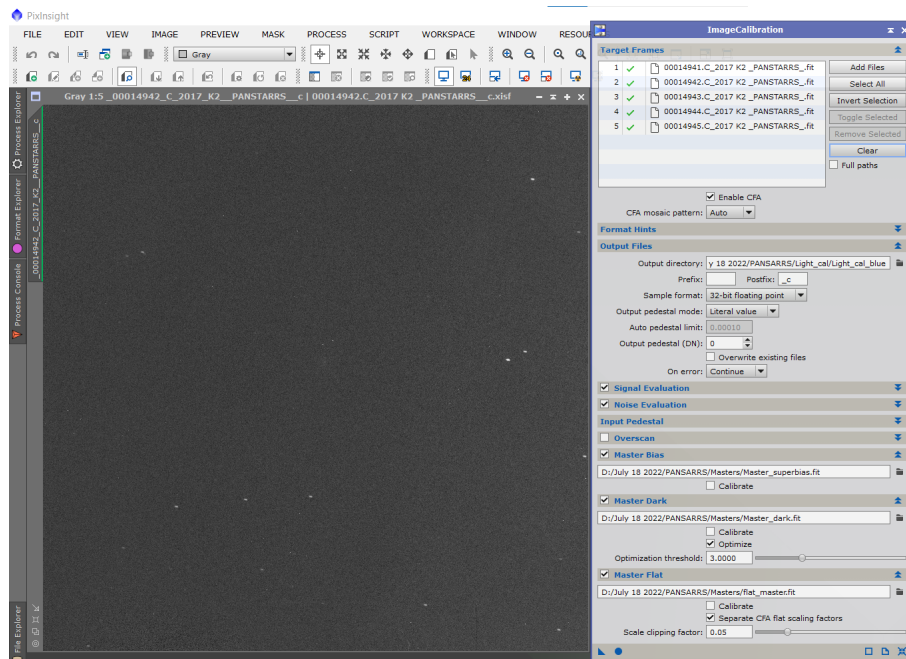


Figure 19: Light calibration of a light frame

Significant changes can be realized after the calibration is applied to the raw images, as viewed in Figure 20. Much of the noises, white strips, dust pattern, and the dark area in the left of the lower edge were effectively removed. The calibrated frames are smoother and the objects (the stars and the central comet) in the frame is easier to detect.

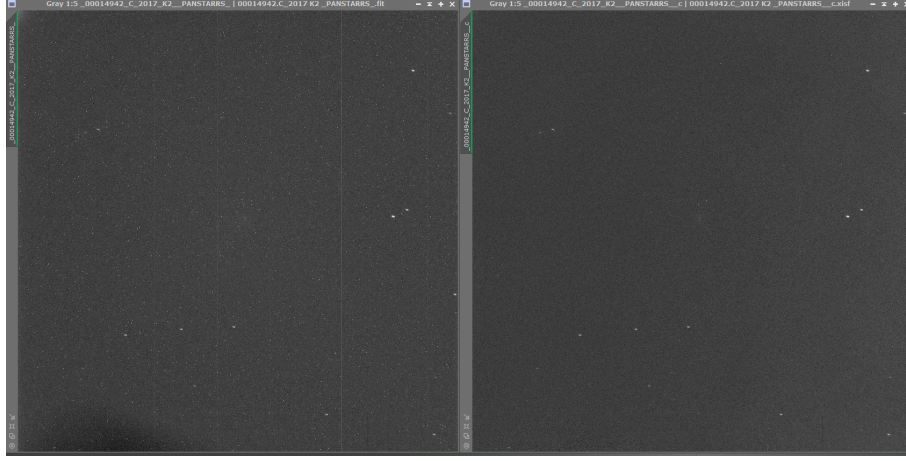


Figure 20: Before and after calibration of a light frame

3.2.2 Cosmetic Correction

Cosmetic Correction is the process of removing the hot pixels in the images that the calibration process did not remove. This step should be done manually, and there are two ways to correct the images: (1) correct the frames individually then integrate and (2) correct the integrated light frame. If done individually, it takes more time but the hot pixels are better removed from each light frame since we can reduce the hot pixels caused by cosmic rays in each frame, and the integration of these frame will produce less noise images. If we correct the integration light frame, it takes less time but the hot pixels are not well treated and still appear in the output frames. Therefore, I chose method (1) for my processing.

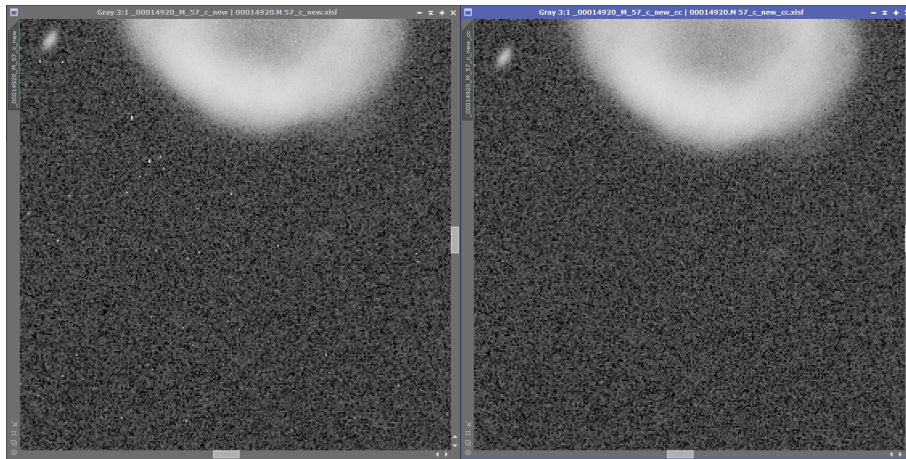


Figure 21: Before (left) and after (right) cosmetic correction

3.2.3 Image Registration

Since the images may have some small differences in the position of the star, we need to align some of the points to obtain the right position for later integration. There are three types of alignment in the ImageRegistration in PixInsight: Comet alignment, Dynamic Alignment, and Star Alignment.

- Comet Alignment: Allow alignment of comet images. Since the comet moves at a different velocity from the background stars, we will need to know the velocity and the direction of the comet so that the software will know how to align the images.
- Dynamic Alignment: Can be applied for any case, in the trade of time-consuming. This is manual alignment, where the user will choose a “source” image and a “target” image, and choose some stars in the source image to tell the target image to align to. The greater the number of stars, the better the registration. The minimum number of stars should be 15.
- Star Alignment: The stars in the images will be automatically recognized by PixInsight, and the software will automatically align them. The software requires a few hundred stars to be able to align.

Since we deal with 2 types of images: nebula images and comet images, we need to approach different methods to align these two. In both of the cases, I use Dynamic Alignment because the number of images is not too large and it is easy to use.

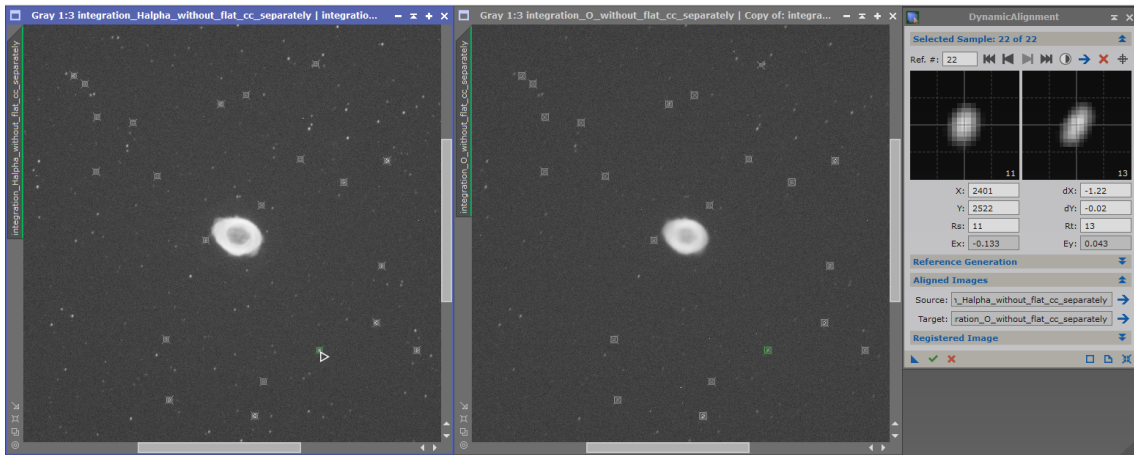


Figure 22: Image registration using Dynamic Alignment

3.2.4 Image Integration

We do the same as before when we do with calibration frames. I did not do anything more special since I have only five light frames for each filter. If there were more light frames, we can work with the subframe weighting formula to choose which frame should weigh more than others.

3.2.5 Cosmic rays

Cosmic rays are very easily spotted in astronomical images. Thao et al. [intro2HL] have calculated the frequency of cosmic rays hit is one every 4 seconds across the 14 cm^2 CCD. Figure 23 is an RGB combination of three images taken at three different times. We can see that there are a lot of cosmic rays hit in each frame, and some are very strong, hence we can hardly get rid of them, especially when they hit right in the center of the object that we observe. The cosmic rays' hits can be got rid of by some integration algorithm like average. However, when the cosmic ray hits, we still lose some information in that frame, and sometimes we need to get rid of the frame so that it does not affect the overall result.

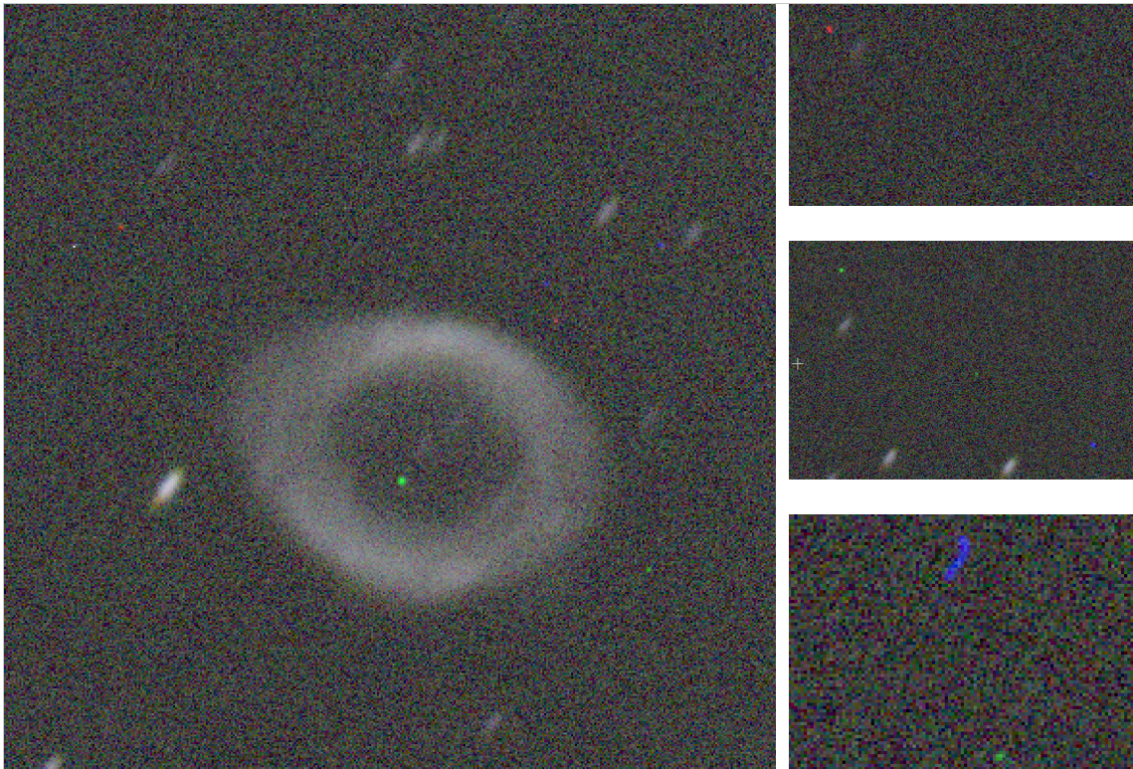


Figure 23: Cosmic rays

4 Results

4.1 M57 image

Messier 57, (also catalogued as Ring Nebula, M57 and NGC 6720), is a planetary nebula lies in the Lyra constellation. Its diameter is about one light-year. [M57'hubble]

- RA: 18h 53m 35.16s
- Dec: 33° 1' 43.2"
- Distance: 0.7 kpc or 2300 light-years
- Apparent Magnitude: 8.8
- Size: 3'48" x 2'24"

With the size of 3'48" x 2'24", the Ring Nebula lies entirely in the frames, and ranges about more than 200 pixels in each direction.

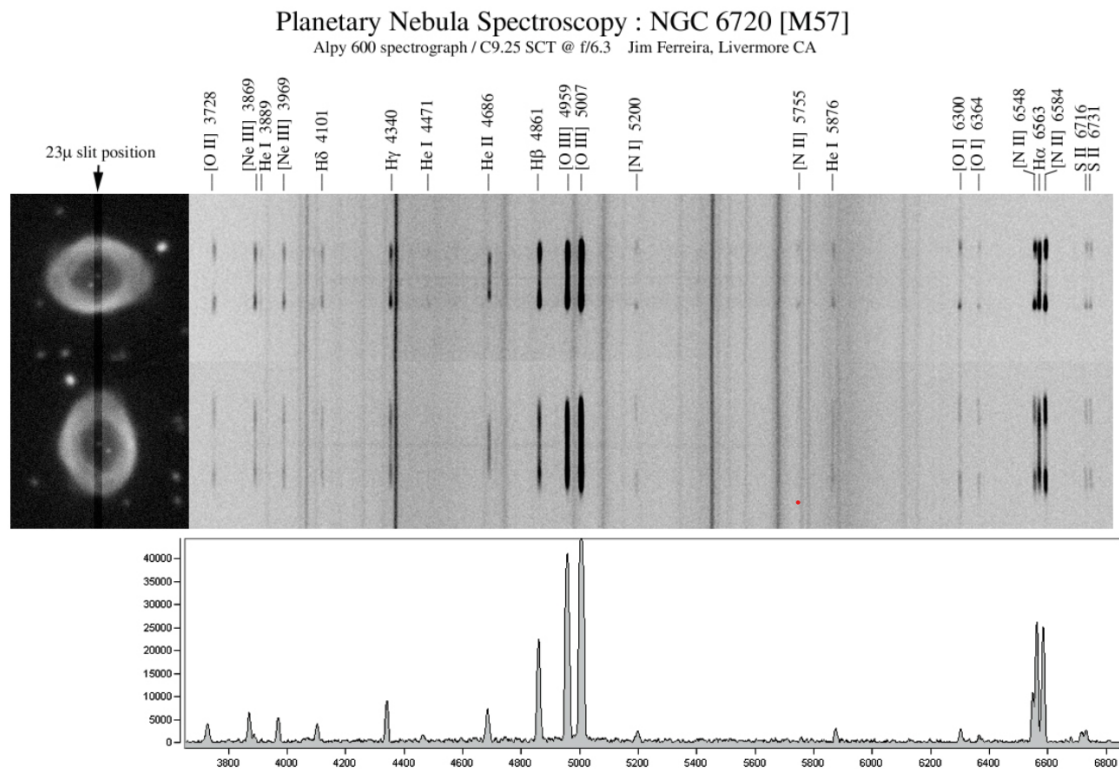


Figure 24: M57 Spectrum, modified from [M57'Spectrum]

From the spectroscopy in Figure 24, we can see that the Ring nebula has a high content of O-III, the Balmer series, and Nitrogen. The content of S-II is less significant and quite

faint in comparison with O-III and H-alpha. This is corresponding to the light frames that we obtain, shown in Figure 25: the Ring nebula in S-II light frames are fainter than in H-alpha and O-III. From the statistics of the pixels in the central region in Figure 26, we can see that the intensity of the S-II is one-fifth of the H-alpha intensity and one-third of the O-III intensity. In Figure 27 and 28, the H-alpha is colored red, the O-III channel is colored green and S-II is colored blue.

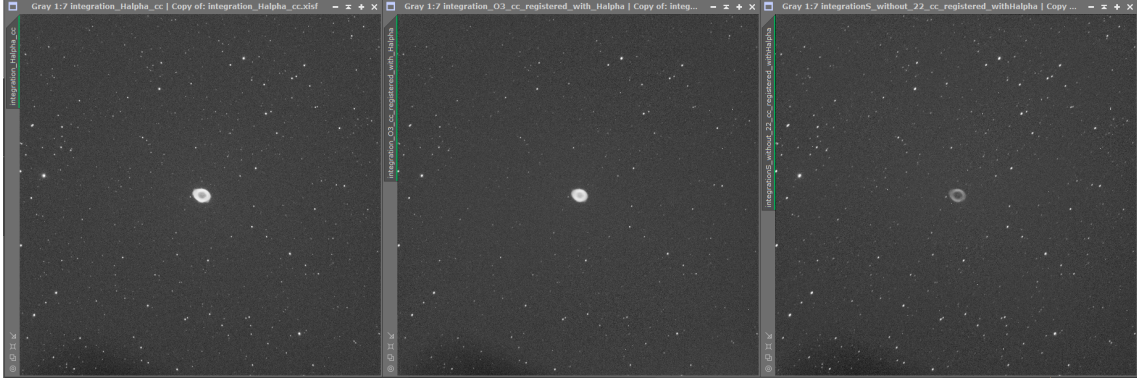


Figure 25: H α , O-III and S-II, respectively

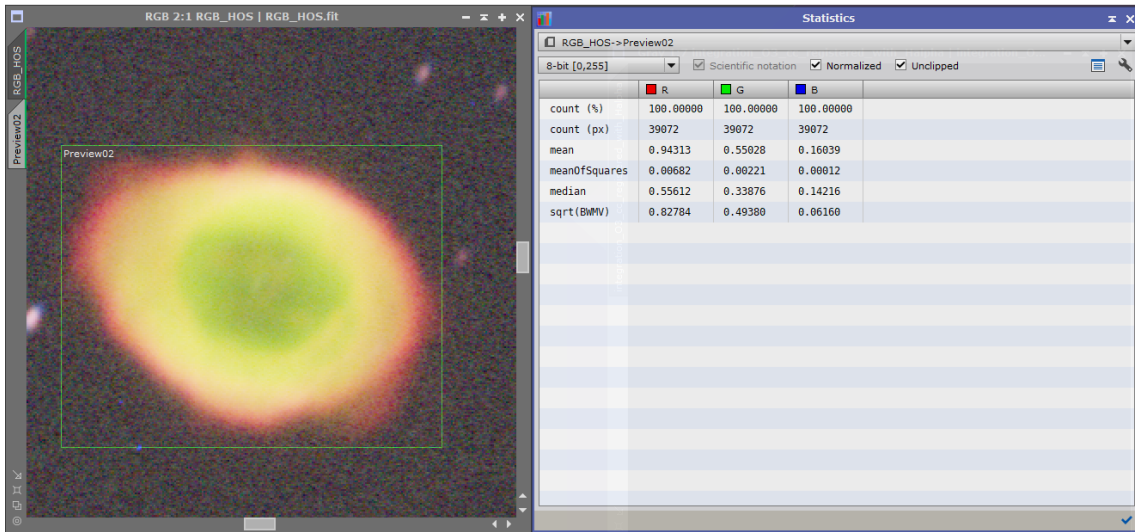


Figure 26: Statistics of the three channel in the central region

By channel combination of the three channels, the intensity of each channel is well observed. H-alpha, the red color, surrounds the center of the ring, meanwhile, the O-III is distributed in all the nebula. The color yellow is a combination of green and red, hence the color region contains both H-alpha and O-III. Since the intensity of the S-II channel is weak in comparison with the other two channels, we can hardly see the color blue or

the white region in the nebula image. The central star of the nebula is also invisible in the images, possibly due to its small size and the quality of the CCD.

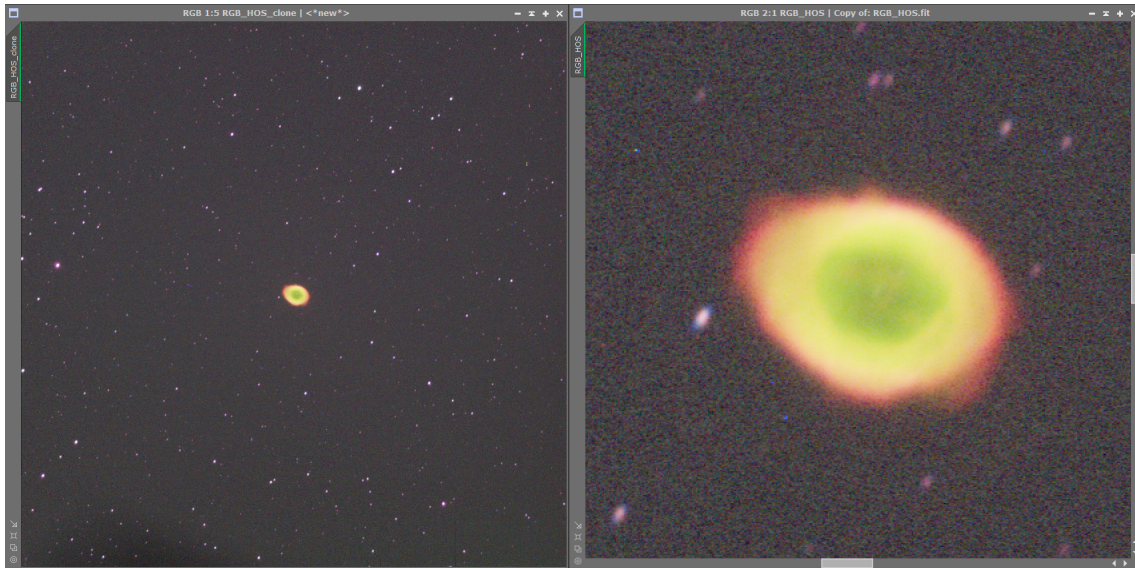


Figure 27: RGB combination of $H\alpha$, O-III and S-II, respectively

The *AssistedColorCalibration* tool in PixInsight aims to modify the weight of each channel in the color combination image by dividing or multiplying the pixel values in each channel. With this tool, we can neutralize the background to see less noise and the pattern of the nebula also be easier to see.

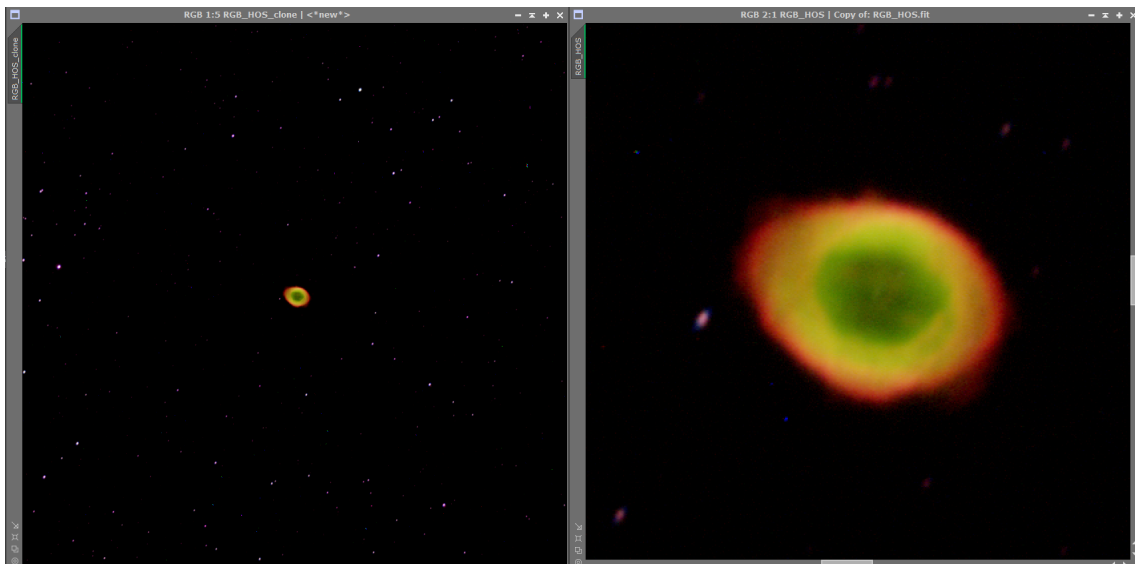


Figure 28: RGB combination of $H\alpha$, O-III and S-II, respectively, color assisted

4.2 C/2017 K2 (PanSTARRS)

This comet was first observed in 2017 when it was 16 AU from the Sun. After a long time of observing and calculating, Małgorzata Królikowska and Piotr A. Dybczyński [K2'orbit'oort] found that this comet is an old comet from the Oort cloud, and its perihelion has been predicted to happen in 2022. The orbit elements of K2 is obtain from [K2'orbit'params'1, K2'orbit'params'2]. The path of K2 in 2022 and 2023 has been visualized by Tom Ruen, which is noted in Appendix 6.3.

Orbit Parameters

[hide]

Osculating Orbital Elements

Epoch 2459173.5 (2020-Nov-20.0) TDB

Reference: [JPL 82](#) (heliocentric IAU76/J2000 ecliptic)

Element	Value	Uncertainty (1-sigma)	Units
e	1.000459046955121	1.715E-6	
a	-3917.572997986876	14.625	au
q	1.798349956189691	5.787E-6	au
i	87.54730878453114	9.2075E-6	deg
node	88.25469020338348	1.4479E-5	deg
peri	236.1781676998013	.00016585	deg
M	-0.00305429918700722	1.7104E-5	deg
tp	2459933.359240336928	.00041146	TDB
period	not defined	not defined	d
n	4.01955918263614E-6	2.2509E-8	deg/d
Q	not defined	not defined	au

Model Parameters

Parameter	Value	Uncertainty (1-sigma)	Units
A1	3.206740092957519E-8	4.948E-9	au/d^2
A2	-3.095825358867455E-8	2.356E-9	au/d^2
ALN	.0408373333128795		
NK	2.6		
NM	2.		
NN	3.		
R0	5.		au

Miscellaneous Details

solution date	2022-Aug-29 11:13:38
# obs. used (total)	223
data-arc span	3392 days (9.29 years)
first obs. used	2013-05-12
last obs. used	2022-08-25
planetary ephem.	DE441
SB-pert. ephem.	SB441-N16
norm. resid. RMS	.41732
source	JPL
producer	Davide Farnocchia
Earth MOID	1.09109 au
Jupiter MOID	1.25595 au
T_jup	0.170
Nongrav. accels. using CO driven g(r)	

Figure 29: C/2017 K2's orbit parameters

Capturing this comet, we use the red, green, and blue filters, to see whether the color of the comet is blue or green, since we have seen several images of this comet and its color varies from green to blue. The movement of the comet is not identical to the sky's movement, hence we could not make use of the field rotation and the guiding tube to track the comet. Therefore, we must take multiple short exposure frames to reduce the trace of the comet. By the time that we were taking the images of this comet, its right ascension is $16^{\circ}49'52.567''$ and its declination is $-05^{\circ}07'07.35''$, corresponding with the altitude of 29.552° and azimuth 256.758° . According to [tracking time], we took 5 frames in each filter, exposure time 20 seconds each frame, and took dark frames in 20 seconds exposure each frame also.

From West to East, in the North half, the sky is contaminated by the city light. Unluckily, at the time that we take the images, the comet was in the West - South - West, as shown in Figure 30, hence we can see the big difference in the red, green, and blue light masters when applying the same Screen Transfer Function (STF), shown in Figure 31.

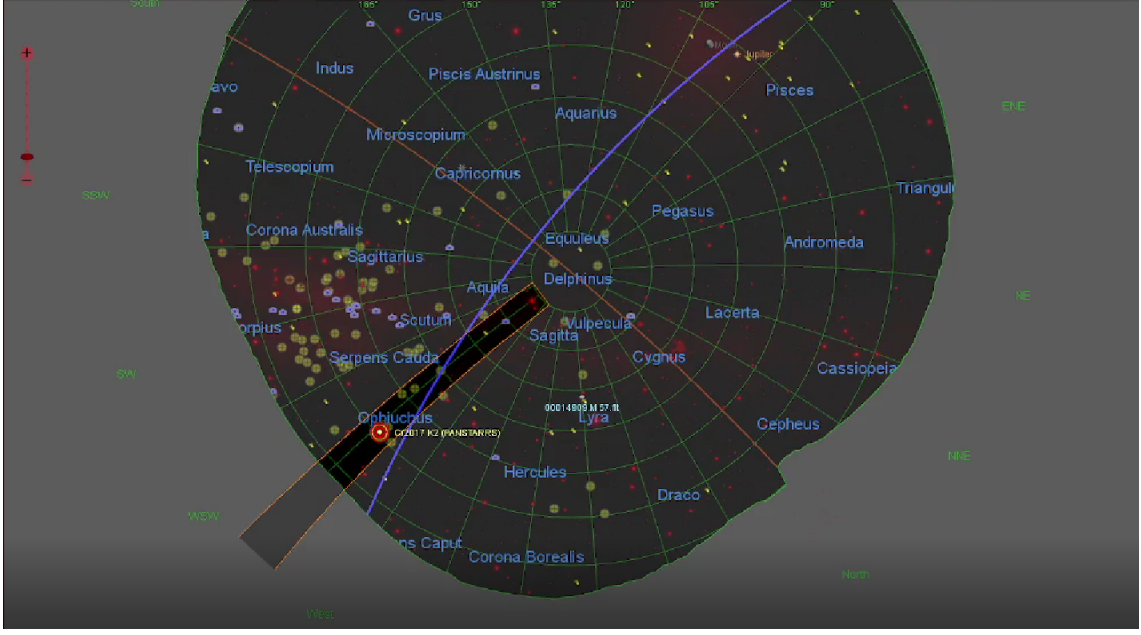


Figure 30: Position of K2 PanSTARRS when the images were taken

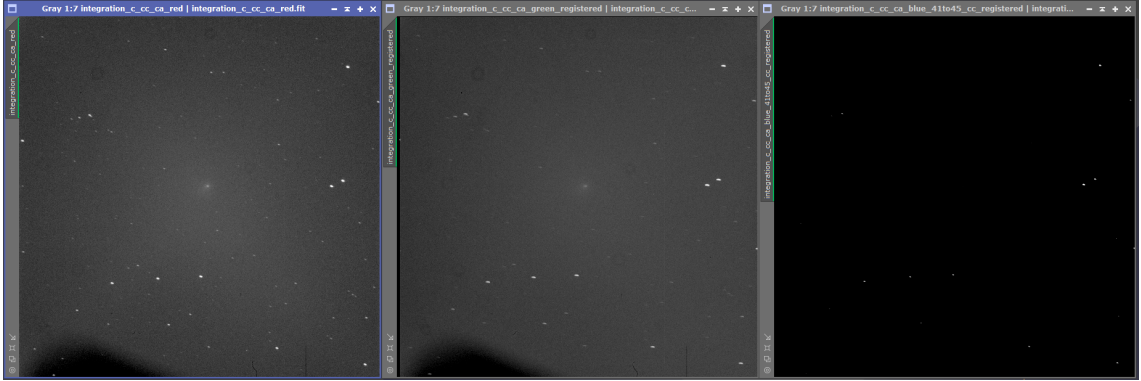


Figure 31: Red, green, and blue master lights, with the same STF

After the channel combination of the three color channels, we obtained the color image of K2, in Figure 32. We can see the yellow color all over the image, due to the lack of blue channel. Hence, we could not answer the question of the color of the K2 and must plan another session to take the image of K2 at different times and locations, to reduce the light pollution.

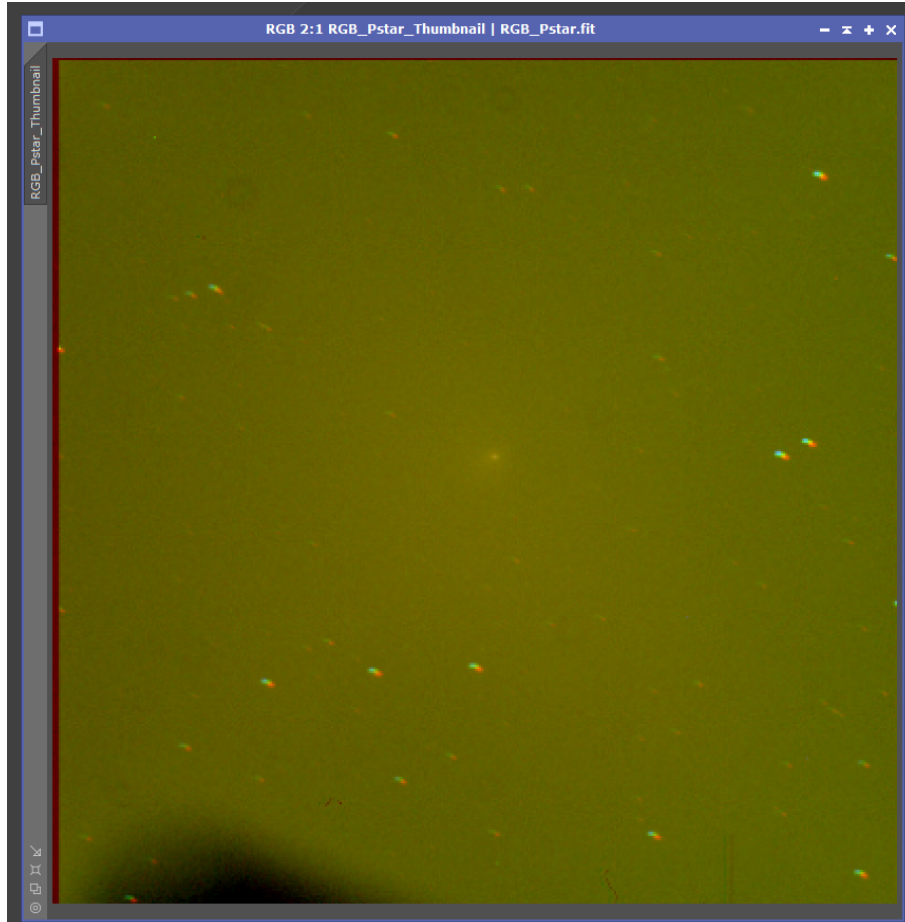


Figure 32: RGB channel combination of C/2017 K2

Even though we took the light frame in a relatively small exposure time, we did not obtain the circular pattern of the stars as well as the comet, but rather slightly inclined ellipses. By evaluating the ellipses of the stars and with the help of automatic ellipse identification, I learn that the ellipse has an average eccentricity of 0.532 and inclination of -6.7 degrees. Considering the highly active region of the comet covered an area of 20×14 pixels in the light frame that we obtain, as depicted in Figure 33, and the FOV of a pixel is 0.000128 degrees, and the distance to Earth of the comet at that time is 270 million kilometers, I roughly estimate that this area has a diameter about 9800 kilometers. This is a far cry from the outer limit that the material of the comet can expand into space (130,000 km) and from the estimated nucleus diameter of this comet (14–80 km) [K2'wiki]. However, this seems to be a reasonable result since the material of the comets will densely cover the comet in a large range, and it matches the images that were taken with other telescopes.

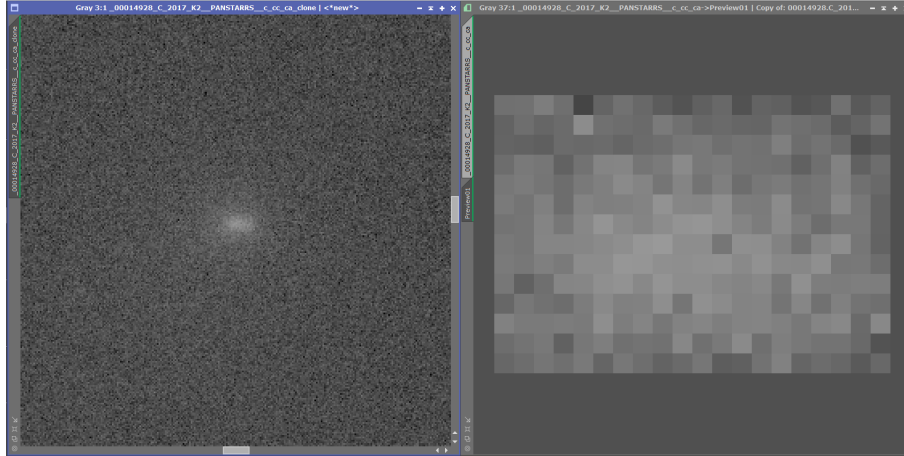


Figure 33: Center of K2

5 Conclusions

During the internship in Nha Trang Observatory, I have the opportunity to operate a large telescope system and utilize its facility, as well as understand the atmospheric effects that act on the imaging procedure. We have taken the images of the Ring Nebula and C/2017 K2 Comet with different filters. The results are incomplete due to short amount of time and limitations of the apparatus. This internship has given me the chance to first practice on the astrophotographs.

6 Appendix

6.1 CCD data sheet

Camera Performance			
Typical Maximum Cooling	50°C below ambient	Dark Current (typical)	<0.02 electrons/pixel/sec at -30°C
Temperature Stability	0.1°C	Cooling Method	Air (Optional liquid)
Digitization Speed	1 MHz, 8 MHz		
Typical System Noise	9 e- RMS @ 1 MHz; 17 e- at 8 MHz	Non-Linearity	<1%
Housing Dimensions	6.2 X 6.2 X 3.8 inches (15.7 X 15.7 X 9.6 cm)	Weight	5.6 lbs (2.5 kg)
Focal Plane to Face Plate	21 mm		
Lens Mounts	Optional Nikon F-mount or Canon EOS mount		
Interface	USB 2.0	Camera Channels	1
Available Shutters	65 mm	Shutter MTBF	1000000
External Triggering	Standard		
Environment	-30°C to 45°C 10% - 90% Relative Humidity		
Power	12V (100-240V AC to 12V DC power supply included). With TEC off: <1A. TEC at 100%: 4.6A. Shutter open: 4A pulse for 100msec. Shutter held open, add 0.22A.		

Figure 34: FLI PL16801 CCD data sheet

6.2 Types of CCD

- Front side illumination: CCDs are illuminated through the electrodes. Electrodes are semi-transparent, but some losses occur.
- Back side illumination: CCDs are illuminated from the back side.
- Front-side illuminated device: relatively low QE, filling factor $\eta < 1$
- Back-side illuminated device (or thinned devices): high QE and physically thinned to about 15 microns, filling factor close to 1.

6.3 Path of K2

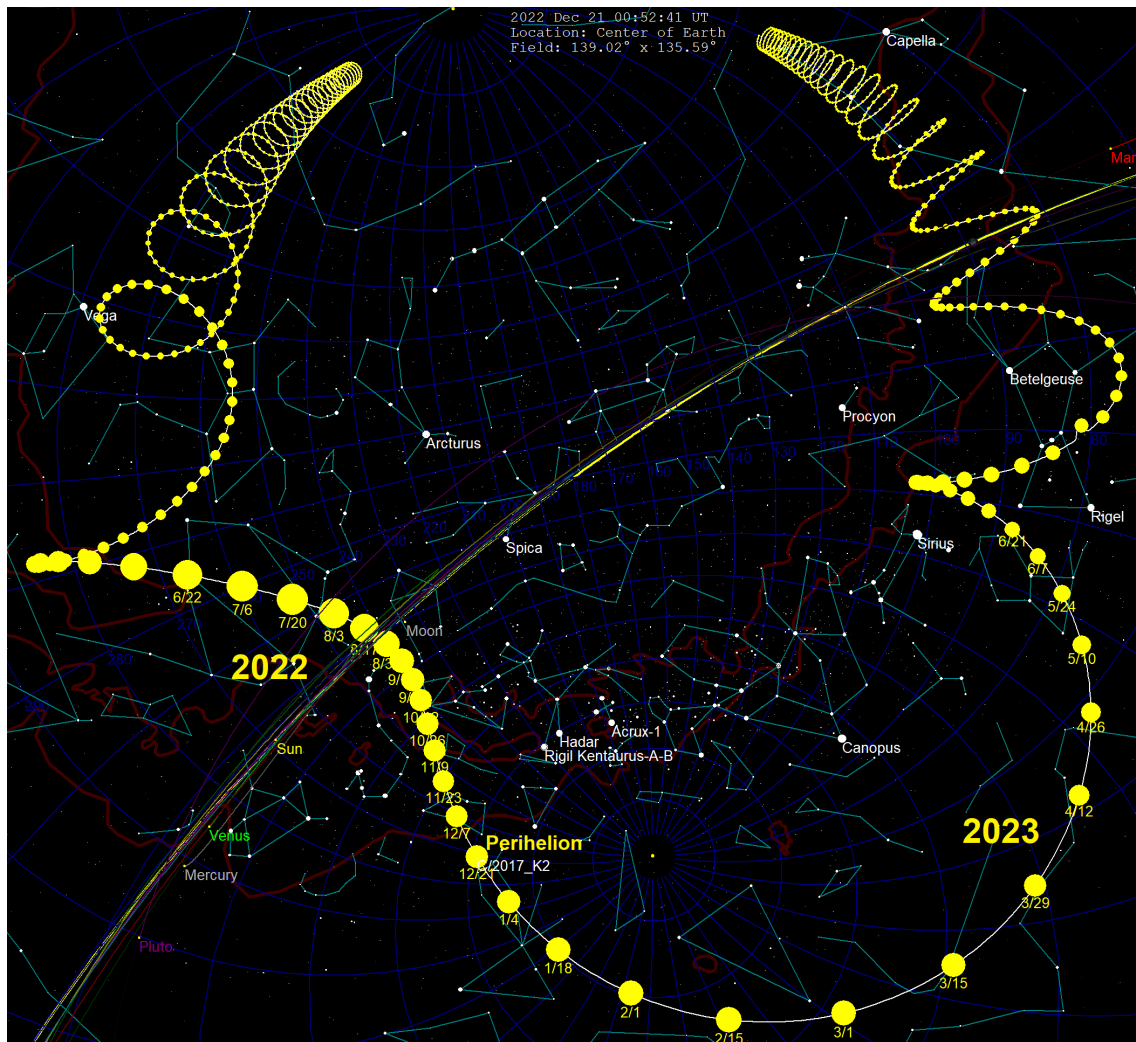


Figure 35: Path of K2 in celestial sphere [K2'path]

6.4 Drying the CCD

Aware of the status of the CCD, we tried to dry it by blowing out the wet air inside the CCD chamber. The idea is to use dry nitrogen flow (the laboratory nitrogen goes through silica gel) and let it blow inside the CCD in one hole, and the wet air inside the CCD will go out the other hole, as depicted in Figure 36.

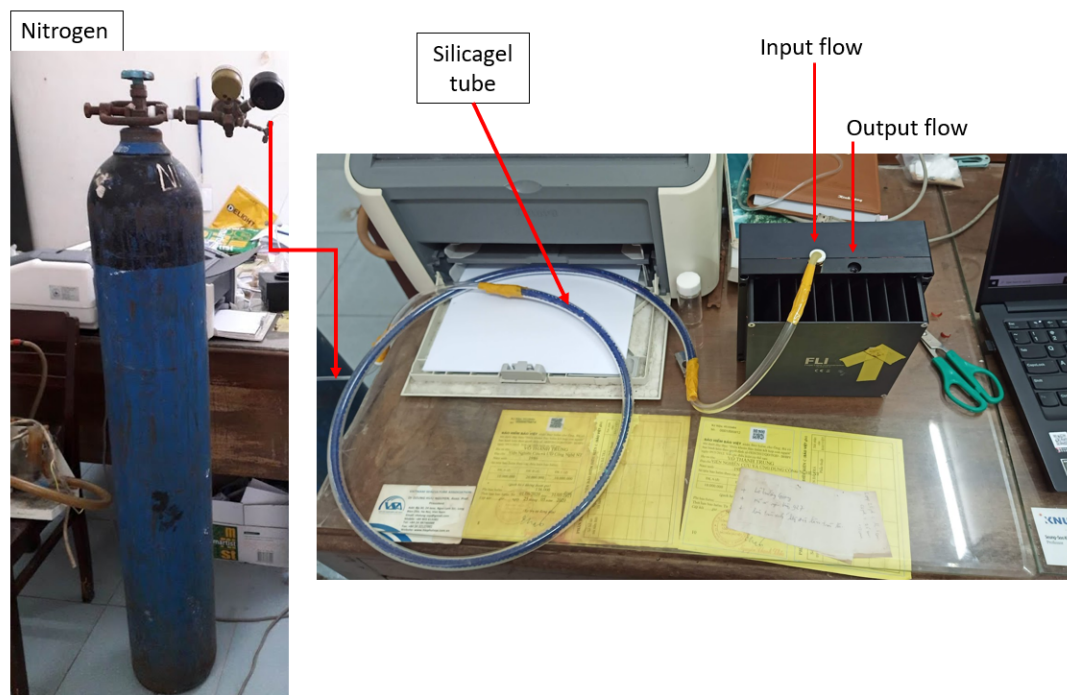


Figure 36: The setup for CCD drying

We let the flow blow in for more than 30 minutes, and then we tested the CCD. But the flat field is still heavily contaminated with the water, and the CCD was not fixed. My assumption is that the CCD array was isolated from the space that we blew the dry air into. In the future, we will send the CCD back to the manufacturer for full maintenance.